

AI-Enabled Water Well Predictor

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Abstract—Access to reliable groundwater is a critical component of sustainable infrastructure, directly aligned with UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure). Traditional well-site assessment relies heavily on manual hydrogeological surveys, which are time-consuming, resource-intensive, and often inaccessible in under-resourced regions. This project proposes an AI-enabled water well predictor that leverages historical groundwater monitoring data, along with associated rainfall and temperature records, to forecast groundwater levels across multiple stations in India. The system uses a multistation spatiotemporal dataset spanning 1994–2025, incorporating groundwater levels, rainfall, and temperature as core features. After preprocessing (handling missing values, feature engineering for seasonal and lagged climate effects), the project applies machine learning regression models — beginning with a Random Forest/Gradient Boosting baseline — to predict groundwater levels at a given location and time. These predictions are further translated into an interpretable well-risk classification (e.g., Safe, Declining, Critical), enabling practical decision-making for well-site viability and water resource planning. The final deliverable includes a lightweight web-based interface where a user can input a region and time period to receive a predicted groundwater level along with a corresponding risk category. By combining time-series climate data with modern machine learning techniques, this project aims to deliver a scalable, low-cost decision-support tool for groundwater management, without dependency on physical sensors or IoT infrastructure — improving automation and real-world usability while keeping the solution accessible and reproducible.

I. INTRODUCTION

Access to reliable groundwater is a critical component of sustainable infrastructure, directly aligned with United Nations Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure). Groundwater accounts for a significant share of water supply in agricultural and semi-arid regions of India, yet traditional well-site assessment continues to rely on manual hydrogeological surveys. These surveys are time-consuming, resource-intensive, and frequently inaccessible in under-resourced regions, creating a gap between the need for reliable water access and the tools available to assess it.

Recent advances in machine learning (ML) have demonstrated strong potential for groundwater level prediction using historical climate and hydrological records, offering a scalable alternative to physics-based hydrological simulation. However, prior work has largely focused on maximizing predictive accuracy without translating model outputs into decision-ready information for non-technical stakeholders.

This paper proposes an AI-enabled water well predictor that forecasts groundwater levels using historical monitoring data, rainfall, and temperature records across multiple stations in India, and further classifies the predicted level into an interpretable risk category (Safe, Declining, or Critical). The system is implemented as a lightweight, sensor-free web application, requiring no additional hardware or IoT infrastructure, making it accessible and reproducible for practical deployment.

II. RELATED WORK

A review of recent literature on machine learning approaches to groundwater level prediction was conducted to inform model selection and feature engineering strategy.

Rohde et al. [1] applied Random Forest regression using satellite-based climate and groundwater data across California, finding that 44% of groundwater-dependent ecosystems showed significant long-term decline, and that ML-based approaches required substantially less calibration than traditional physical hydrological models. Davari et al. [2] compared XGBoost, Random Forest, and Support Vector Machine (SVM) across five hydrogeological zones in Iran, reporting XGBoost as the strongest performer ($R^2 = 0.848$, RMSE = 1.554 m), while SVM exhibited overfitting. Jesse et al. [3] conducted a PRISMA-guided systematic review of 223 peer-reviewed articles, confirming that Random Forest and its hybrid variants consistently outperform ANN and SVM baselines, and identified that most ML groundwater models remain “black-box” systems with limited interpretability for policy and decision-making contexts. Sarkar et al. [4] and Pham et al. [5] further reinforced the strength of tree-based ensemble methods

(Random Forest, Random Tree) across independent regional datasets in India and Bangladesh, respectively.

A parallel body of work has explored deep learning approaches. Chen et al. [6] and Thakur et al. [8] applied LSTM-based architectures, noting that data quality and missing-value imputation often matter more than model architecture, and that deep learning models require substantially more historical data and computational resources than tree-based alternatives. Khan et al. [7] and Feng et al. [10] compared classical, machine learning, and deep learning approaches, concluding that no single model architecture dominates universally and that deep learning offers benefit primarily when long, dense time-series data is available. Notably, Ali et al. [9], reviewing 91 deep learning studies, identified a substantial gap between academic research and real-world deployment, finding that most published models are never translated into usable, decision-ready tools.

Across this literature, two consistent gaps emerge: (i) limited interpretability of model outputs for non-expert stakeholders, and (ii) a disconnect between research prototypes and deployable applications. The present work addresses both gaps directly through an interpretable risk classification layer and a deployed web interface.

III. METHODOLOGY

A. Dataset

The dataset used in this study consists of multistation groundwater level records for India spanning 1994–2025, comprising 416,952 daily observations across 6,440 monitoring stations. Each record includes a groundwater level reading, rainfall, average/maximum/minimum temperature at 2m, station latitude and longitude, well depth, aquifer type, and encoded state/district identifiers.

B. Data Cleaning and Preprocessing

Approximately 299 stations exhibited physically implausible extreme readings (e.g., values below -2000 or above 2000), likely attributable to reference-datum inconsistencies across monitoring stations. These outliers were addressed by clipping the target variable to the 1st and 99th percentile range rather than removing affected stations entirely, preserving data volume while limiting the influence of erroneous readings.

C. Feature Engineering

Two engineered features were found to be critical for prediction accuracy, consistent with prior literature [3]: (i) a lag feature representing each station’s previous available groundwater reading, and (ii) a rolling three-reading average of rainfall per station. Records lacking a valid lag value (the first reading per station) were excluded from training.

D. Model Training

Two ensemble regression models were trained and compared: Random Forest and XGBoost. Data was split into training (80%) and held-out test (20%) subsets. Both models were trained on the following feature set: month, rainfall,

average/maximum/minimum temperature, latitude, longitude, well depth, aquifer type, state, district, previous groundwater level, and rolling rainfall average.

E. Risk Classification

To translate continuous groundwater level predictions into decision-ready output, a per-station risk classification scheme was defined using historical percentile thresholds:

- **Safe:** predicted level at or below the station’s historical 50th percentile
- **Declining:** predicted level between the 50th and 80th percentile
- **Critical:** predicted level above the 80th percentile

Thresholds are computed independently per station to account for regional variation in baseline groundwater depth.

F. Deployment

The trained model is deployed via a lightweight Streamlit web application, allowing a user to select a monitoring station and input climate conditions to receive a predicted groundwater level alongside its corresponding risk classification, without requiring any physical sensors or IoT infrastructure.

IV. RESULTS AND DISCUSSION

A. Model Performance

Table I summarizes the performance of both models on the held-out test set.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	R ²	RMSE	MAE
Random Forest	0.975	2.374	1.153
XGBoost	0.977	2.281	1.128

XGBoost marginally outperformed Random Forest across all three metrics and was selected as the final deployed model.

B. Feature Importance

Feature importance analysis revealed that the lag feature (previous groundwater reading) accounted for approximately 94% of the model’s predictive weight, substantially outweighing all climate and static features combined. This finding is consistent with the systematic review by Jesse et al. [3], which identified lagged groundwater levels as the most predictive feature across the ML groundwater literature, and reflects the strong temporal autocorrelation inherent in groundwater level time series.

C. Risk Distribution Across Stations

Applying the per-station risk classification scheme to the most recent reading at each of the 6,108 monitoring stations reveals a distribution skewed toward stable conditions, as shown in Figure 1. Approximately 51% of stations were classified as Safe, 24% as Declining, and 24% as Critical. This distribution reflects the classification scheme’s design, where thresholds are computed per station from historical data, and

provides a system-wide snapshot of groundwater risk that can support prioritization of intervention across a large number of monitoring locations without requiring manual review of each station's raw time series.

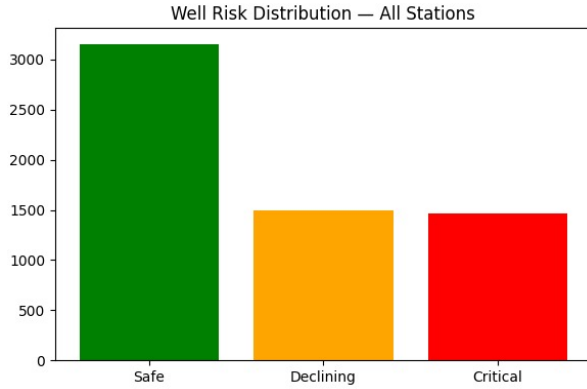


Fig. 1. Well risk distribution across all monitoring stations, based on the most recent available reading per station.

D. Discussion

The high R^2 achieved (0.977) confirms that ensemble tree-based models are well-suited to this prediction task, consistent with findings from Davari et al. [2] and Pham et al. [5]. However, the dominance of the lag feature suggests that the model's performance is driven primarily by temporal persistence rather than climate-driven forecasting; this is an expected characteristic of groundwater systems, which change gradually, but implies that the model's forecasting horizon should be interpreted with caution for long-range predictions where no recent reading is available.

The risk classification layer directly addresses the interpretability gap identified in [3] and the deployment gap identified in [9], translating a numerical prediction into a decision-ready Safe/Declining/Critical category without requiring end-users to interpret raw model output.

V. CONCLUSION AND FUTURE WORK

This paper presented an AI-enabled water well predictor combining ensemble machine learning (Random Forest and XGBoost) with an interpretable risk classification layer, deployed as a sensor-free web application. Using a real-world dataset of 416,952 groundwater readings across 6,440 Indian monitoring stations, the proposed XGBoost model achieved an R^2 of 0.977, demonstrating strong predictive performance despite significant real-world data quality challenges, including reference-datum inconsistencies and irregular reading intervals.

Future work will focus on: (i) adopting a time-based train/test split to more rigorously evaluate forecasting performance under realistic deployment conditions, (ii) incorporating explainability techniques such as SHAP to provide feature-level justification alongside each prediction, (iii) extending the risk classification scheme with expert-validated thresholds

rather than purely statistical percentiles, and (iv) evaluating model generalizability across additional geographic regions beyond the current dataset.

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